SYS-CONTROL

PREDICTIVE MAINTENANCE & THERMAL EQUIPMENT ANALYTICS

Proof of Concept (POC) Architecture, Model Calibration, and Technical Specification Report

Document Purpose: This technical specification details the architecture, design choices, physical and statistical methodologies, and evaluation outcomes of the SYS-CONTROL Predictive Maintenance system. It outlines the machine learning classifiers trained for mechanical failures (rotary machinery) and the first-principles thermodynamic calculations implemented for static equipment (heat exchangers).

Document Version:	1.0.0 (Production Blueprint)
Status:	Approved POC / Technical Reference
Published Date:	July 09, 2026
Target Systems:	Rotary Fleet (Type L, M, H) & Shell-and-Tube Heat Exchangers
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1. Executive Summary

This project establishes a unified predictive maintenance framework that combines machine learning classifiers with first-principles thermodynamic analytics. By leveraging the industry-standard AI4I 2020 Predictive Maintenance dataset and classic heat transfer relations, the system is split into two operating domains:

• **Domain A - Rotary Machinery (ML-Based):** Focuses on predicting sudden mechanical failures in high, medium, and low-cost production tools. Using advanced machine learning (XGBoost and Random Forest), the system predicts overall failure probability and isolates 5 specific failure modes (Tool Wear, Heat Dissipation, Power, Overstrain, and Random Failures) to recommend direct engineering interventions.

• **Domain B - Heat Exchangers (Physics-Based):** Evaluates thermal degradation and fouling resistance using thermodynamic relationships. Using an Effectiveness-NTU approach, it rates the health of static thermal equipment, ranks them by criticality, and schedules maintenance before units reach warning or critical limits.

• **Operational Flexibility:** Models are calibrated with Platt Scaling and support dual operating profiles: a *Balanced Mode* which maximizes the F1-Score (operational efficiency) and a *Safety-First Mode* which guarantees catching at least 90% of failures (91.18% empirical recall) to protect high-consequence assets.

Key Outcome: *The primary XGBoost classifier achieved a high Area Under the ROC Curve (ROC-AUC) of* ***0.9776****. Through decision threshold calibration, the system caught* ***91.18%*** *of failures in Safety-First mode, offering a robust safeguard against unscheduled plant downtime.*

2. Problem Statement & Domain Context

Industrial plants lose millions annually due to unscheduled maintenance. Traditional reactive 'run-to-failure' strategies are costly, while conservative time-based preventative maintenance schedules replace expensive parts too early, leading to wasted useful life. This POC solves these problems for two equipment paradigms:

2.1 Rotary Equipment & Mechanical Failure Modes

The system monitors physical variables (Air/Process Temperature, Rotational Speed, Torque, Tool Wear) of production machines. There are 5 distinct failure mechanisms modeled:

• **Tool Wear Failure (TWF):** Tool wear exceeds physical limits (between 200 and 240 min), causing mechanical failure.

• **Heat Dissipation Failure (HDF):** Occurs if the delta between process and air temperature is too narrow (< 8.6 K) while rotational speed is low (< 1380 rpm), preventing effective thermal dissipation.

• **Power Failure (PWF):** The actual mechanical power (Torque * Speed) drops below 3500 W or exceeds 9000 W, causing motor stalls or structural overloading.

• **Overstrain Failure (OSF):** Combined stress of high torque and tool wear. The product of torque and wear exceeds standard structural thresholds (e.g., > 11,000 Nm·min for Type L, > 12,000 for Type M, > 13,000 for Type H).

• **Random Failures (RNF):** Stochastic process anomalies not fully explained by physical features, occurring with an empirical probability of 0.1%.

2.2 Static Thermal Equipment & Fouling

Heat exchangers transfer energy between hot and cold fluid streams. Over time, particulate accumulation, scaling, and chemical deposition create a solid layer of fouling resistance (R_f). This fouling degrades the overall heat transfer coefficient (U) and reduces thermal effectiveness. Unmonitored fouling leads to thermal pinch bottlenecks, forcing premature plant shutdowns for manual cleaning.

3. System Architecture

The SYS-CONTROL system is designed as a decoupled, modern multi-tier application. The architecture separates computation-heavy machine learning inference and thermal calculations from the user interface.

Component / Layer	Technology / Implementation	Description / Role
Frontend Dashboard	React, Vite, CSS, Chart.js / Plotly	An operator console displaying real-time equipment statuses, simulation controls, degradation forecasts, and historical csv upload panels.
Backend REST API	FastAPI, Uvicorn, Pydantic	Exposes routes in main.py for model inference, time-to-failure calculations, and heat exchanger thermal audits. Stores registry database in JSON.
ML Inference Pipeline	XGBoost, Scikit-Learn, Pickle	Loads serialized XGBoost and RF models. Transforms raw features dynamically and predicts probability scores.
TTF Forecasting	Python, NumPy, ttf_engine.py	Runs an iterative forward simulation using drift rates to forecast remaining useful life (RUL) and suggest interventions.
Static Thermal Analytics	Python, heat_exchanger_analyzer.py	Implements first-principles thermodynamics (NTU-Effectiveness) to rate heat exchanger health and rank units.

3.1 Key Code Files in the Repository

• **main.py:** FastAPI app. Initializes, loads models and scalers, exposes routes like /predict, /simulate-ttf, and /hx/fleet-status.

• **1_train_model.py:** Primary ML training pipeline. Preprocesses training sets, performs feature engineering, applies SMOTE, trains classifiers, calibrates thresholds, and writes evaluations.

• **ttf_engine.py:** RUL simulator. Predicts the month of failure by forward-stepping parameters and checks model predictions under normal vs. intervened states.

• **heat_exchanger_analyzer.py:** Thermodynamic heat transfer model. Evaluates fouling resistance, effectiveness, and calculates estimated days until cleaning is required.

• **machine_registry.py / machine_registry.json:** Equipment database management layer. Houses registration schema, metadata, and operational parameters for all tracked physical units.

4. Machine Failure Predictive Modeling (ML Pipeline)

Machine learning models are trained on the AI4I 2020 dataset (10,000 original rows). Due to the heavy imbalance of the dataset (~3.39% raw failure rate), specific feature engineering and sampling steps were required.

4.1 Feature Engineering (Physics-Informed)

Raw physical measurements are transformed into higher-order features to capture thermodynamic and mechanical stress relations:

1. **Temperature Delta (temp_delta)**: Tracks heat accumulation. Process Temp minus Air Temp [K].
2. **Mechanical Power (power_W)**: Formulated from torque and speed: *Power = Torque * Speed * (2 * pi / 60)* [Watts].
3. **Torque-Wear Interaction (torque_x_wear)**: Represents cumulative stress on the cutting tools: *Torque * Tool Wear*.
4. **Tool Wear Percentage (wear_pct)**: Ratio of current tool wear to maximum tool wear (253.0 min).
5. **Stress Indicators (high_torque, high_wear)**: Binary indicators for torque exceeding 55 Nm or wear exceeding 200 min.

4.2 Handling Severe Class Imbalance

To prevent classifiers from ignoring the minority class (failures) and predicting 'no failure' universally, the training pipeline applies two techniques:

• **SMOTE (Synthetic Minority Over-sampling Technique):** Synthesizes new minority instances (failures) in the training set using k-nearest neighbors (k=5). This balances the training target (y_train_sm) to a 50/50 ratio, allowing the model to learn the decision boundaries of failure states effectively.

• **Scale Position Weight (scale_pos_weight):** For individual failure mode sub-classifiers (TWF, HDF, PWF, OSF, RNF), SMOTE is not applied. Instead, a dynamic class weight multiplier is calculated as *n_neg / n_pos*. This penalizes false negatives on specific failure modes severely during training.

4.3 Model Probability Calibration (Platt Scaling)

Standard tree-based models, especially when trained with synthetic oversampling (SMOTE) or high scale_pos_weight, output distorted, non-calibrated probability estimates. While the ranking of risks remains correct, the absolute value of the output (e.g. 0.82) does not represent the true likelihood of failure. To correct this, the production XGBoost model uses **Platt Scaling** via Scikit-Learn's *CalibratedClassifierCV* with a 5-fold cross-validation loop.

Calibration maps raw classification scores into true empirical probability estimates. If the calibrated model predicts an 80% chance of failure, it means that empirically, 80 out of 100 machines with those exact physical indicators will fail. This calibration is crucial for operators when defining alarm thresholds.

5. Model Performance & Calibration

The models were evaluated on a hold-out test set (ai4i2020_test_real.csv, 2,000 rows) with zero data leakage. We compared XGBoost and Random Forest. Both models were calibrated and evaluated across two operating profiles:

1. **Balanced Mode**: Threshold calibrated to maximize F1-score. Suitable for routine operations.
2. **Safety-First Mode**: Threshold calibrated to guarantee a minimum 90% Recall, catching critical anomalies at the expense of a higher false-alarm rate.

Model & Profile	Thresh.	F1-Score	Recall (RUL)	Precision	Conf. Matrix (TN, FP, FN, TP)
XGBoost (Balanced)	0.82	0.8281	77.94% (53/68)	88.33%	[[1925, 7], [15, 53]]
XGBoost (Safety-First)	0.08	0.5210	91.18% (62/68)	36.47%	[[1824, 108], [6, 62]]
Random Forest (Balanced)	0.74	0.7705	69.12% (47/68)	87.04%	[[1925, 7], [21, 47]]
Random Forest (Safety-First)	0.25	0.5536	91.18% (62/68)	39.74%	[[1838, 94], [6, 62]]

5.1 Model Selection & Rationale

XGBoost (ROC-AUC: 0.9776) is selected as the primary online inference model over **Random Forest (ROC-AUC: 0.9712)** due to superior metrics in Balanced Mode (F1-score of 0.8281 vs 0.7705, and Recall of 77.94% vs 69.12%). XGBoost's gradient boosting algorithm captures complex non-linear combinations of rotational speed, torque, and tool wear more effectively. Furthermore, in Safety-First mode, setting the decision threshold to 0.08 allows the XGBoost model to achieve 91.18% recall while limiting false alarms (108 false alarms out of 2000 total samples).

6. Time-to-Failure (TTF) Forecasting Engine

The TTF Engine (implemented in ttf_engine.py) provides actionable maintenance schedules by projecting sensor parameters month-by-month using linear operational degradation drift rates.

6.1 Default Degradation Drift Rates

Based on fleet averages, the engine applies the following parameter changes per operating month:

Parameter	Monthly Drift Rate	Physical Interpretation
Tool Wear [min]	+12.0 min	Normal cutting tool material degradation.
Rotational Speed [rpm]	-25.0 rpm	Motor efficiency loss and friction increase.
Torque [Nm]	+1.2 Nm	Increased mechanical resistance due to wear.
Air Temperature [K]	+0.05 K	Ambient seasonal drift approximation.
Process Temperature [K]	+0.10 K	Internal operating friction heat accumulation.

6.2 Forecasting Loop & Operational Interventions

At each simulation step (Month *m*), the engine calculates physical engineered features (such as *power_W* and *torque_x_wear*) and evaluates the probability of failure using the calibrated XGBoost model. If the probability crosses the active threshold, the engine logs a failure, identifies the primary failure mode (e.g. HDF, OSF), and simulates recommended interventions. The engine tests three intervention actions:

• **Part Replacement:** Resets the tool wear parameter to 0 min, extending the TTF substantially.

• **Load Reduction:** Reduces operating torque by 15% and rotational speed by 10% to lower physical strain on the machine.

• **Ambient Cooling:** Simulates the installation of external cooling systems, lowering ambient air temperature by 5 K, which resolves Heat Dissipation (HDF) risks.

7. Heat Exchanger Thermal Analytics (First-Principles)

Unlike rotary machinery, static heat exchangers do not require machine learning. Instead, they are monitored using first-principles thermodynamics based on the **Effectiveness-NTU (Number of Transfer Units)** method.

7.1 Mathematical Model & Equations

The system evaluates the thermal performance using the following formulation:

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1. Heat Capacity Rates:
C_shell = mass_flow_shell * Cp_shell | C_tube = mass_flow_tube * Cp_tube
C_min = min(C_shell, C_tube) | C_max = max(C_shell, C_tube)
C_ratio = C_min / C_max

2. Actual Heat Duty (Q_actual):
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Q_actual = mass_flow_shell * Cp_shell * (Temp_shell_in - Temp_shell_out) [kW]

3. Maximum Theoretical Heat Duty (Q_max):
Q_max = C_min * (Temp_hot_in - Temp_cold_in) [kW]

4. Thermal Effectiveness (epsilon):
epsilon = Q_actual / Q_max

5. Number of Transfer Units (NTU) & Heat Transfer Coefficient (U):
Using the counter-flow relation (based on configuration):
NTU = (1 / (C_ratio - 1)) * ln((epsilon - 1) / (epsilon * C_ratio - 1))
U_actual = NTU * C_min / Area [W/m^2 K]

6. Thermal Fouling Resistance (R_f):
R_f = (1 / U_actual) - (1 / U_design) [m^2 K/W]
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7.2 Industry Standard Fouling Limits (TEMA)

Fouling resistance (R_f) threshold limits are established using standard Tubular Exchanger Manufacturers Association (TEMA) and Heat Exchanger Design Handbook (HEDH) parameters. When a unit's calculated fouling resistance exceeds these limits, or when thermal effectiveness drops, maintenance alerts are generated:

Fluid Type	Fouling Limit (R_f)	Typical Industrial Application
Cooling Water	0.000176 m^2 K/W	Utility cooling loops, cooling towers.
Process Liquid	0.000352 m^2 K/W	Organic chemical streams, hydrocarbon fractions.
Crude Oil	0.000528 m^2 K/W	Refinery preheater trains (heavy deposition).
Steam	0.000088 m^2 K/W	Reboilers and utility steam heaters (condensation).
Refrigerant	0.000176 m^2 K/W	Chiller evaporators and condensers.

7.3 Degradation Rating & Days-to-Clean Estimation

Each heat exchanger is assigned a health rating based on design effectiveness degradation:
• **Normal:** Effectiveness >= 80% of design value.
• **Warning:** Effectiveness drops below 80% of design, or R_f exceeds the TEMA limit.
• **Critical:** Effectiveness drops below 65% of design.

The engine calculates **Days-to-Clean** by dividing the remaining allowable fouling increment by a default linear accumulation rate of *5e-7 m^2 K/W per day*. This allows operators to schedule pre-emptive tube cleaning campaigns during planned turnarounds.

8. Technical Recommendations & Deployment Plan

Based on the results of the Proof of Concept, the following steps are recommended for production deployment:

• **Model Deployment:** Deploy the calibrated XGBoost model as a microservice using the FastAPI backend. Use the calibrated threshold of 0.82 for general operations to minimize false alarms, and 0.08 for critical, unredundant rotary machinery to prevent catastrophic mechanical failure.

• **Edge Data Integration:** Connect the FastAPI backend to the plant's SCADA or historian system (e.g. OSIsoft PI) to feed sensor streams (temperatures, speed, torque) into the /predict endpoint at 5-minute intervals.

• **Continuous Retraining:** Establish a closed-loop system where operator feedback (confirming true/false alarms) is recorded in the registry and used to retrain the classifiers quarterly, improving model precision over time.

• **Thermal Monitoring:** Implement the first-principles Heat Exchanger analytics using automatic thermal data collected from inlet/outlet thermocouples and orifice plate flowmeters.

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